Code No: 114CN **R13**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech II Year II Semester Examinations, September/October - 2023
COMPUTER ORGANIZATION
(Computer Science and Engineering)

Time: 3 Hours **Max. Marks: 75**

Note: i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

PART - A

(25 Marks)

1.a)	What is the need of overflow flag?	[2]
b)	What is the purpose of memory?	[3]
c)	What is the need of I/O bus?	[2]
d)	What is asynchronous data transfer?	[3]
e)	What is the need of ROM in main memory?	[2]
f)	Illustrate address translation in virtual memory.	[3]
g)	What is max mode of 8086?	[2]
h)	What is the content of flag register?	[3]
i)	List iteration control instructions.	[2]
j)	What is the difference between CBW and CWD instructions?	[3]

PART - B

(50 Marks)

2. Describe the basic components of computer and their organizations. [10]

OR

3. With suitable examples, explain conditional branch instructions. [10]

4. Make a comparison of programmed I/O with interrupt initiated. [10]

OR

5. Demonstrate the steps in communication between CPU and I/O processor. [10]

6. Explain the importance of content addressable memory and its match logic. [10]

OR

7. Differentiate between associative and set-associative mapping techniques of cache memory. [10]

8. Discuss the purpose of the following pins in 8086:
NMI, INTA, LOCK, ALE, DEN [10]

OR

9. Illustrate various addressing modes supported by 8086 with examples. [10]

10. Demonstrate evaluation of arithmetic expressions in 8086 using the following expression:
24 * 12 + 10 / 2 – 36 *9 [10]

OR

11. Write an assembly language program to print prime numbers till 100. [10]

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